

ECE4010J Probabilistic Methods in Engineering

Extra Exercises and Solutions

Exercise 1 (10 pts) Adult onset diabetes is known to be highly genetically determined. A study was done comparing frequencies of a particular allele in a sample of diabetics and a sample of nondiabetics. The data are shown in the following table.

	Diabetic	Normal
Bb or bb	15	5
BB	35	45

Use a chi-square test for homogeneity to see whether the frequencies of the alleles are different in the two groups at a significance level of 0.05.

Solution. Let H_0 be that the genotype proportions are the same for the diabetic and normal groups. The row totals are 20 and 80, the column totals are 50 and 50, and the grand total is 100. Thus

$$E_{ij} = \frac{(\text{row total})(\text{column total})}{100},$$

so the expected counts are

	Diabetic	Normal
Bb or bb	10	10
BB	40	40

The degrees of freedom are

$$(2 - 1)(2 - 1) = 1.$$

The chi-square statistic is

$$\chi^2 = \frac{(15 - 10)^2}{10} + \frac{(5 - 10)^2}{10} + \frac{(35 - 40)^2}{40} + \frac{(45 - 40)^2}{40} = 6.25.$$

Since $\chi_{0.05,1}^2 = 3.84$ and $6.25 > 3.84$, we reject H_0 at the 0.05 level. Therefore, the allele frequencies are significantly different in the two groups.

Exercise 2 (30 pts) Let $X_1, \dots, X_n$ be a random sample from a distribution with density

$$f_\theta(x) = \frac{x}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right), \quad x > 0, \quad \theta > 0.$$

(i) **(10 pts)** Find the maximum likelihood estimator $\hat{\theta}_{\text{ML}}$ of θ .

Solution. The likelihood is

$$L(\theta) = \prod_{i=1}^n \frac{x_i}{\theta^2} \exp\left(-\frac{x_i^2}{2\theta^2}\right) = \left(\prod_{i=1}^n x_i\right) \theta^{-2n} \exp\left(-\frac{1}{2\theta^2} \sum_{i=1}^n x_i^2\right).$$

Hence

$$\ell(\theta) = \sum_{i=1}^n \ln x_i - 2n \ln \theta - \frac{1}{2\theta^2} \sum_{i=1}^n x_i^2.$$

Differentiating,

$$\ell'(\theta) = -\frac{2n}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^n x_i^2.$$

Setting $\ell'(\theta) = 0$ gives

$$\hat{\theta}_{\text{ML}}^2 = \frac{1}{2n} \sum_{i=1}^n X_i^2, \quad \hat{\theta}_{\text{ML}} = \sqrt{\frac{1}{2n} \sum_{i=1}^n X_i^2}.$$

(ii) **(10 pts)** Find the method of moments estimator $\hat{\theta}_{\text{MM}}$ of θ .

Solution. First compute

$$E[X] = \int_0^\infty x f_\theta(x) dx = \int_0^\infty \frac{x^2}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right) dx.$$

Using $y = x/\theta$,

$$E[X] = \theta \int_0^\infty y^2 e^{-y^2/2} dy = \theta \sqrt{\frac{\pi}{2}}.$$

The method of moments sets $\bar{X} = E[X]$, so

$$\bar{X} = \theta \sqrt{\frac{\pi}{2}}, \quad \hat{\theta}_{\text{MM}} = \sqrt{\frac{2}{\pi}} \bar{X} \approx 0.7979 \bar{X}.$$

(iii) **(10 pts)** Find $\text{Var}(\hat{\theta}_{\text{MM}})$ in terms of θ .

Solution. We need $\text{Var}(X)$. Since

$$E[X^2] = \int_0^\infty x^2 f_\theta(x) dx = \int_0^\infty \frac{x^3}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right) dx = 2\theta^2,$$

we have

$$\text{Var}(X) = 2\theta^2 - \left(\theta \sqrt{\frac{\pi}{2}}\right)^2 = \frac{4-\pi}{2} \theta^2.$$

Therefore,

$$\text{Var}(\hat{\theta}_{\text{MM}}) = \text{Var}\left(\sqrt{\frac{2}{\pi}} \bar{X}\right) = \frac{2}{\pi} \cdot \frac{\text{Var}(X)}{n} = \frac{4-\pi}{\pi n} \theta^2 \approx \frac{0.2732}{n} \theta^2.$$

Exercise 3 (10 pts) A certain response time (in ms) for two materials are measured, resulting data

Material A	251	255	258	257	250	251	254	250	248
Material B	250	253	249	256	259	252	260	251	

Use the Wilcoxon rank-sum test to test $H_0 : \mu_A \leq \mu_B$ at level $\alpha = 0.05$.

Solution. Rank all observations together, using average ranks for ties. The rank sum for Material B is

$$W_B = 4 + 10 + 2 + 13 + 16 + 9 + 17 + 7 = 78.$$

Here $m = 8$ for Material B and $n = 9$ for Material A. Under H_0 ,

$$E[W_B] = \frac{m(m+n+1)}{2} = \frac{8(18)}{2} = 72,$$

and

$$\text{Var}(W_B) = \frac{mn(m+n+1)}{12} = \frac{8 \cdot 9 \cdot 18}{12} = 108.$$

Using the normal approximation,

$$Z = \frac{W_B - E[W_B]}{\sqrt{\text{Var}(W_B)}} = \frac{78 - 72}{\sqrt{108}} = 0.5774.$$

For this left-tailed alternative, we reject only for sufficiently small values of Z . Since $0.5774 > -1.645$, we do not reject H_0 at the 0.05 level.

Exercise 4 (25 pts) A group of 5 patients treated with medicine A is of weight

42, 39, 38, 60, 41

kg. A second group of 7 patients from the same hospital treated with medicine B is of weight

38, 42, 56, 64, 68, 69, 62

kg. Assume that the samples are drawn from normal distributions. We want to know whether there is any difference between medicines.

- (i) **(10 pts)** Test $H_0 : \sigma_A^2 = \sigma_B^2$ at $\alpha = 0.2$ level to decide whether pooling is appropriate.

Solution. The sample variances are

$$s_A^2 = 82.5, \quad s_B^2 = 154.3333.$$

The test statistic is

$$F = \frac{s_A^2}{s_B^2} = \frac{82.5}{154.3333} = 0.5346, \quad F \sim F_{4,6} \quad \text{under } H_0.$$

Equivalently,

$$\frac{s_B^2}{s_A^2} = 1.8707, \quad F \sim F_{6,4}.$$

For the two-sided test with $\alpha = 0.2$, the relevant upper-tail cutoff is $F_{0.1,6,4} = 4.010$. Since $1.8707 < 4.010$, we do not reject H_0 . Thus pooling is appropriate.

- (ii) **(10 pts)** Depending on the result from (i), test $H_0 : \mu_A = \mu_B$ at $\alpha = 0.05$ level.

Solution. Since pooling is appropriate, use the pooled two-sample t test. We have

$$\bar{X}_A = 44, \quad \bar{X}_B = 57,$$

and

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2} = \frac{4(82.5) + 6(154.3333)}{10} = 125.6.$$

Therefore,

$$T = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{s_p^2(1/n_A + 1/n_B)}} = \frac{44 - 57}{\sqrt{125.6(1/5 + 1/7)}} = -1.9810.$$

The critical value is $t_{0.025,10} = 2.228$. Since

$$|T| = 1.9810 < 2.228,$$

we do not reject H_0 at the 0.05 level.

(iii) **(5 pts)** Find a 90% confidence interval on $\mu_A - \mu_B$.

Solution. For a 90% confidence interval, $\alpha = 0.1$ and $t_{0.05,10} = 1.812$. Thus

$$(\bar{X}_A - \bar{X}_B) \pm t_{0.05,10} \sqrt{s_p^2 \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}$$

gives

$$-13 \pm 1.812 \sqrt{125.6 \left(\frac{1}{5} + \frac{1}{7} \right)} = -13 \pm 11.8908.$$

Hence the 90% confidence interval is

$$[-24.891, -1.109].$$

Exercise 5 (25 pts) Given the following data

$$\begin{array}{c|cccccccccc} x & 39 & 65 & 62 & 90 & 82 & 75 & 25 & 98 & 36 & 78 \\ y & 47 & 53 & 58 & 86 & 62 & 68 & 60 & 91 & 51 & 84 \end{array}$$

Consider the linear regression model $y = \beta_0 + \beta_1 x$. The following summary statistics are given:

$$\sum_{i=1}^n x_i = 650, \quad \sum_{i=1}^n y_i = 660, \quad \sum_{i=1}^n x_i^2 = 47648, \quad \sum_{i=1}^n y_i^2 = 45784.$$

(i) **(5 pts)** Estimate the slope and the intercept of the linear model.

Solution. Here $n = 10$, $\bar{x} = 65$, and $\bar{y} = 66$. Also,

$$S_{xx} = \sum x_i^2 - n\bar{x}^2 = 47648 - 10(65)^2 = 5398,$$

$$S_{yy} = \sum y_i^2 - n\bar{y}^2 = 45784 - 10(66)^2 = 2224.$$

From the data,

$$\sum x_i y_i = 45604,$$

so

$$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 45604 - 10(65)(66) = 2704.$$

Therefore,

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{2704}{5398} = 0.50093,$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 66 - 0.50093(65) = 33.4398.$$

- (ii) **(6 pts)** Find a 95% confidence interval for the slope and the intercept.

Solution. First,

$$\text{SSE} = S_{yy} - \hat{\beta}_1 S_{xy} = 2224 - 0.50093(2704) = 869.4954.$$

Thus

$$s^2 = \frac{\text{SSE}}{n-2} = \frac{869.4954}{8} = 108.6869, \quad s = 10.4253.$$

With $t_{0.025,8} = 2.306$, the 95% confidence interval for β_1 is

$$\hat{\beta}_1 \pm t_{0.025,8} \frac{s}{\sqrt{S_{xx}}} = 0.50093 \pm 2.306 \frac{10.4253}{\sqrt{5398}},$$

so

$$\beta_1 : [0.1737, 0.8281].$$

For the intercept,

$$\hat{\beta}_0 \pm t_{0.025,8} s \sqrt{\frac{\sum x_i^2}{n S_{xx}}} = 33.4398 \pm 2.306(10.4253) \sqrt{\frac{47648}{10(5398)}}.$$

Hence

$$\beta_0 : [10.8530, 56.0265].$$

- (iii) **(4 pts)** Find a 90% prediction interval for $Y \mid (x = 60)$.

Solution. The fitted value at $x_0 = 60$ is

$$\hat{y}(60) = \hat{\beta}_0 + \hat{\beta}_1(60) = 33.4398 + 0.50093(60) = 63.4954.$$

For a 90% prediction interval, use $t_{0.05,8} = 1.860$. Thus

$$\hat{y}(60) \pm t_{0.05,8} s \sqrt{1 + \frac{1}{n} + \frac{(60 - \bar{x})^2}{S_{xx}}}$$

equals

$$63.4954 \pm 1.860(10.4253) \sqrt{1 + \frac{1}{10} + \frac{(60 - 65)^2}{5398}} = 63.4954 \pm 20.3803.$$

Therefore, the 90% prediction interval is

$$[43.115, 83.876].$$

- (iv) **(10 pts)** Estimate the Pearson correlation coefficient ρ and test $H_0 : \rho = 0$ at $\alpha = 0.025$.

Solution. The sample correlation is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{2704}{\sqrt{5398(2224)}} = 0.78041.$$

To test $H_0 : \rho = 0$, use

$$T = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.78041\sqrt{8}}{\sqrt{1-0.78041^2}} = 3.5302.$$

Under H_0 , $T \sim t_8$. For a two-sided test with $\alpha = 0.025$, the critical value is $t_{0.0125,8}$; in particular, $t_{0.0125,8} < t_{0.005,8} = 3.355$. Since

$$|T| = 3.5302 > 3.355,$$

we reject H_0 at the $\alpha = 0.025$ level. There is significant evidence of nonzero linear correlation.